

Introduction to Pharmaceutical Analysis

Science

May, 2026

Introduction to Pharmaceutical Analysis (A11410)

Short Title: Intro to Pharma Analysis
Faculty Science and Computing
Department Science
Cluster Analytical Science
Author EOWENS
Credits 10

Level: Introductory

Description of Module / Aims

The aim of the module is to introduce the student to the basic principles of chemical analysis for the identification, purification and separation of organic molecules. The techniques discussed will be techniques that are routinely used in the QC/QA or research laboratories of a pharmaceutical company.

Programmes

PHAR-0008 Higher Certificate in Science in Good Manufacturing Practice and Technology 1 / 2 / M
(WD_SGMPT_C)

Indicative Content

- Identify common organic functional groups
- Name simple organic compounds including isomers, saturated and unsaturated compounds
- Introduction to spectroscopic techniques: Ultraviolet (UV), Infrared (IR)
- Introduction to chromatographic techniques: High Performance Liquid Chromatography (HPLC), Gas Chromatography (GC)
- Use of Karl Fischer titration for moisture determination in quality control
- Melting point determinations for sample purity
- Thermogravimetric and Differential Scanning Calorimetry (DSC) analysis for sample composition and purity determination
- Refractometry for sample composition and purity determination
- Viscosity and specific gravity for sample composition and purity determination

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify the organic functional groups in a pharmaceutical compound, name basic organic compounds using IUPAC nomenclature.
2. Describe the basic components and basic principles of UV/Vis, IR, HPLC and GC instrumentation.
3. Record an IR spectra of a solid and a liquid sample and identify characteristic absorption peaks of pharmaceutical compounds.
4. Record a UV/Vis spectrum and a calibration graph. Using the calibration graph determine the concentration of a sample and its molar extinction coefficient.
5. Construct a calibration graph of peak area against concentration for a compound using HPLC and GC; use the calibration graph to determine the concentration of a sample; separate a simple two component mixture using HPLC and GC.
6. Determine the mp, tg, decomposition temperature, viscosity, specific gravity, water content and refraction index of a sample for composition and purity comparison..
7. Communicate by written means the results of practical investigations.

Learning and Teaching Methods

- Lectures.
- Practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>In-Class Assessment</i>	50%	1,2
<i>Practical</i>	50%	3,4,5,6,7

Assessment Criteria

- <40%:** No understanding of the basic principles of the module content. Inadequate practical skills in the associated laboratories.
- 40%-49%:** Has a superficial knowledge of the module content. Has a basic level of competency and skills in the associated laboratories.
- 50%-59%:** Shows a reasonable level of understanding of the various principles discussed in the module. Displays good practical skills in the associated laboratories.
- 60%-69%:** Shows a good knowledge of the areas discussed in the module as well as an ability to reason and analyse problems related to the area. Has a high level of skills in the associated laboratories.
- 70%-100%:** All the above to an excellent level, in addition to excellent reasoning skills and focused well thought out arguments. Engages in debate in class and has a high level of efficiency in the associated laboratories.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		6
Tutorial		6
Independent Learning		258

Essential Material

- "Introduction to Pharmaceutical Analysis." www.wit.ie/moodle
- Skoog, D.A., D.M. West and F.J. Holler. _Fundamentals of Analytical Chemistry_. 9. USA: Mary Finch, 2014.

Requested Resources

- Room Type: Chemistry Lab